# Lorin Urbantat

MSc Computer Science, ETH Zurich · Machine Learning & Interactive Systems

lorin.urbantat@gmail.com | LinkedIn | GitHub | bruol.me

## EDUCATION

**ETH Zurich, MSc Computer Science** 2025 - 2027
- Expected Graduation: 2027
- Major: Data Management Systems; Minor: Machine Learning
- **Information Security Lab (5.75)**, Computational Semantics for Natural Language Processing (5.75), Advanced Formal Language Theory (5.5)

**ETH Zurich, BSc Computer Science** 2021 - 2024
- Graduated August 2024, GPA 5.04 (6.0 – highest)
- Parallel Programming (5.75), Numerical Methods (5.75), Analysis (5.5), **Visual Computing (5.75)**, Applied Computer Architecture (6), Human Computer Interaction (5.25)

## WORK EXPERIENCE

**Information Security Lab Teaching Assistant, ETH Zurich (part-time)** 2026 - Present
- Test lab exercises and host exercise sessions

**Software Engineer, ETH Chair of Education Systems (part-time)** 2025 - Present
- Sole engineer for a research-focused RAG chatbot serving 1,000 users. Built the application and document processing pipeline, including OCR and image descriptions, using Next.js, Convex and Microsoft Foundry.

**Software Engineer, ETH Game Technology Center (part-time)** 2023 - 2025
- Morph Tales Game, Unity 3D
- Developed features for an interactive Unity 3D game and worked on generative AI in game characters.
- Built a game analytics pipeline using Firebase, BigQuery and Looker Studio.

**Coach, mint & pepper (part-time)** 2025
- STEM and AI tutor for high school students

**Calculus Teaching Assistant, ETH Zurich (part-time)** 2023
- Host exercise classes, 15 - 30 students

## SELECTED PROJECTS & RESEARCH EXPERIENCE

**TLDR - Speech-based Lecture Summarizer, VIScon Hackathon** 2023
- Used OpenAI Whisper and language model APIs to turn lecture recordings into searchable summaries.

**SCION Control Plane Algorithms Test Environment, Bachelor Thesis** 2024
- Built a Docker-based test environment and traffic generator to evaluate algorithms under configurable network topologies and changing link conditions.

**Unikraft on Raspberry Pi 5, ETH Semester Project** 2026
- Ported Unikraft to Raspberry Pi 5, implementing AArch64 boot, memory initialization, interrupts and timers; developed a cache-safe kernel reload mechanism with OpenOCD and GDB.

**Flateroo, Flat Sharing Platform** 2024 - 2025
- Built a React Native application with real-time chat and video streaming; implemented video transcoding, HLS delivery and a PostgreSQL backend.

## SKILLS & INTERESTS

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| Languages | German (native), English (fluent), French (DELF B1) |
| Technologies | Python, PyTorch, C, C++, Unity 3D, Machine Learning, RAG, SQL, TypeScript, React, Git, Linux, Docker |
| Interests | Biking, Sports, Cooking, Dancing, Singing |